

ADVANCED JAVA PROGRAMMING LAB

(Course Code: 23UPCSC1L06)

**A programming laboratory record submitted to Periyar University, Salem
In partial fulfilment of the requirements for the degree of**

MASTER OF COMPUTER APPLICATIONS

By

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DEPARTMENT OF COMPUTER SCIENCE

PERIYAR UNIVERSITY

State University - NAAC 'A++' Grade - NIRF Rank 94

State Public University Rank 40 - SDG Institutions Rank Band: 11-50

SALEM – 636 011

(OCTOBER - 2025)

CERTIFICATE

This is to certify that the Programming Laboratory entitled
“**ADVANCED JAVA PROGRAMMING LAB (23UPCSC1L06)**” is a
Bonafide record work done by Mr. /Ms. _____
Register No: _____ as
partial fulfilment of the requirements for the degree of
Master of Computer Applications, in the Department of Computer Science,
Periyar University, Salem, during the Academic Year 2025-2026.

Staff In-charge

Head of the Department

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Internal Examiner

External Examiner

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SOURCE CODE:

Student.java

```
import java.lang.*;
import java.io.*;

interface Exam
{
    void percent_cal();
}

class Student {
    String name;
    int roll_no, mark1, mark2;
    Student(String n, int r, int m1,int m2) {
        name = n; roll_no = r; mark1 = m1; mark2 = m2;
    }
    void display()
    {
        System.out.println("====" + "\n" + "Student Details" + "\n" + "====" + "\n");
        System.out.println("Name of Student: " + name);
        System.out.println("Roll No. of Student: " + roll_no);
        System.out.println("Marks of Subject 1: " + mark1);
        System.out.println("Marks of Subject 2: " + mark2);
    }

    public static void main(String[] args) {
        Result R = new Result("Mr.X", 12, 93,84);
        R.display();
        R.percent_cal();
    }
}
```

```
class Result extends Student implements Exam {  
    Result(String n, int r, int m1, int m2) {  
        super(n, r, m1, m2);  
    }  
  
    public void percent_cal() {  
        int total = (mark1 + mark2);  
        float percent = (total * 100) / 200;  
        System.out.println("Percentage: " + percent + "%");  
    }  
  
    void display() {  
        super.display();  
    }  
}
```

OUTPUT:

```
Student Details
====

Name of Student: Mr.X
Roll No. of Student: 12
Marks of Subject 1: 93
Marks of Subject 2: 84
Percentage: 88.0%
```

Source Code:

Exception.java

```
class exception {  
    public static void main(String args[]) {  
        try {  
            int a = 30, b = 0; int c = a / b;  
            System.out.println("Result = " + c);  
        } catch (ArithmeticException e) {  
            System.out.println("Can't divide a number by 0");  
        }  
        try {  
            int num = Integer.parseInt("This is a Number..!");  
            System.out.println(num);  
        } catch (NumberFormatException e) {  
            System.out.println("Number format exception");  
        }  
        try {  
            int a[] = new int[5];  
            a[6] = 9;  
        } catch (ArrayIndexOutOfBoundsException e) {  
            System.out.println("Array Index is Out Of Bounds");  
        }  
        try {  
            Object[] objArray = new Integer[10];  
            objArray[0] = "String";  
        } catch (ArrayStoreException e) {  
            System.out.println("ArrayStoreException: " + e.getMessage());  
        }  
    }  
}
```

```
try {  
    String a = null;  
    System.out.println(a.charAt(0));  
} catch (NullPointerException e) {  
    System.out.println("NullPointerException: " + e.getMessage());  
}  
}  
}
```


OUTPUT:

```
_e42e8491/bin exception  
Can't divide a number by 0  
Number format exception  
Array Index is Out Of Bounds  
ArrayStoreException: java.lang.String  
NullPointerException: null
```

Source Code:

AccessModifier.java

```
class Employee{
    public String name;
    protected int employeeId;
    private double salary;

    public Employee(String name,int employeeId,double salary){
        this.name=name;    this.employeeId=employeeId;    this.salary=salary;
    }

    public void displayDetails(){
        System.out.println("Name:"+name);
        System.out.println("Employee ID:"+employeeId);
        System.out.println("Salary:$"+salary);
    }
    protected void calculateBonus(){
        double bonus=salary*0.1;
        System.out.println("Bonus:$"+bonus);
    }
}

class Manager extends Employee{
    private String department;

    public Manager(String name,int employeeId,double salary,String department){
        super(name,employeeId,salary);
        this.department=department;
    }
}
```

```
public void displayDetails(){
    super.displayDetails();
    System.out.println("Department:"+department);
}
}

class Staff extends Employee{
    private int workingHours;
    public Staff(String name,int employeeId,double salary,int workingHours){
        super(name,employeeId,salary);
        this.workingHours=workingHours;
    }
    public void displayDetails(){
        super.displayDetails();
        System.out.println("Working Hours:"+workingHours);
    }
}

public class AccessModifier{
    public static void main(String[] args){
        Manager manager=new Manager("Mr.X",101,50000.0,"Sales");
        Staff staff=new Staff("Mr.Y",202,60000.0,50);
        System.out.println("Manager Details:");
        manager.displayDetails();
        manager.calculateBonus();

        System.out.println();
        System.out.println("Staff Details:");
        staff.displayDetails();
        staff.calculateBonus();
    }
}
```

OUTPUT:

Manager Details:

Name:Mr.X

Employee ID:101

Salary:\$50000.0

Department:Sales

Bonus:\$5000.0

Staff Details:

Name:Mr.Y

Employee ID:202

Salary:\$60000.0

Working Hours:50

Bonus:\$6000.0

Source Code:

MetricConversionApp.java

```
import javax.swing.*;
import java.awt.*;

public class MetricConversionApp extends JFrame {
    private JLabel cmLabel;
    private JTextField cmTextField;
    private JLabel inchLabel;
    private JTextField inchTextField;
    private JButton convertButton;

    public MetricConversionApp() {
        setTitle("Metric Conversion");
        setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
        setSize(300, 150);
        setLayout(new FlowLayout());

        cmLabel = new JLabel("Centimeters:");
        cmTextField = new JTextField(10);
        inchLabel = new JLabel("Inches:");
        inchTextField = new JTextField(10);
        convertButton = new JButton("Convert");

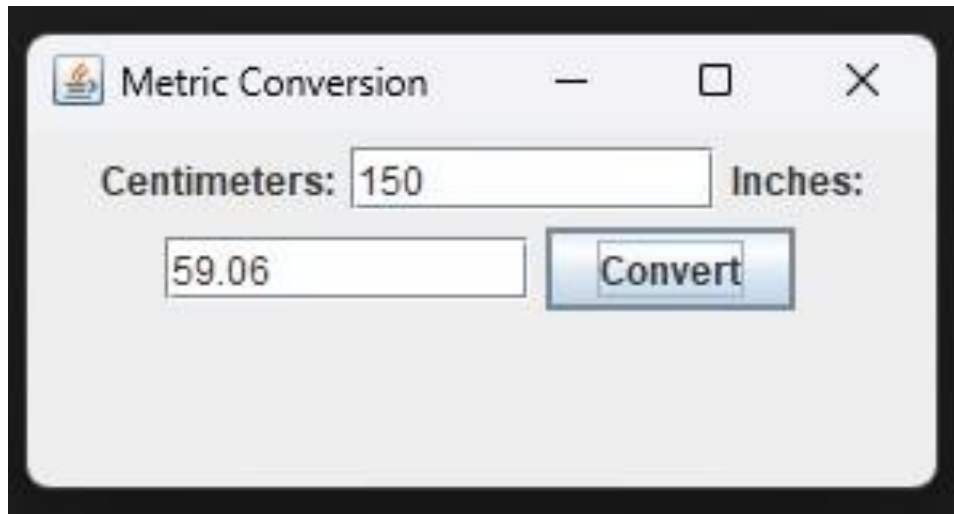
        convertButton.addActionListener(e -> {
            try {
                double centimeters = Double.parseDouble(cmTextField.getText());
                double inches = centimeters / 2.54;
                inchTextField.setText(String.format("%.2f", inches));
            } catch (NumberFormatException ex) {
```

```
        JOptionPane.showMessageDialog(null, "Invalid input. Please enter a valid number.");
    }
});
```

```
add(cmLabel);
add(cmTextField);
add(inchLabel);
add(inchTextField);
add(convertButton);
}
```

```
public static void main(String[] args) {
    SwingUtilities.invokeLater(() -> {
        MetricConversionApp app = new MetricConversionApp();
        app.getRootPane().setDefaultButton(app.convertButton);
        app.setVisible(true);
    });
}
}
```

OUTPUT:



Source Code:

Calc.java

```
import javax.swing.*;
import java.awt.*;
import java.awt.event.ActionEvent;
import java.awt.event.ActionListener;

public class calc extends JFrame implements ActionListener {
    private JTextField textField;
    private JButton[] buttons;
    private String input = "";
    private double num1, num2, result;
    private char operator;
    public calc() {
        setTitle("Simple Calculator");
        setSize(300, 400);
        setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
        setLayout(new BorderLayout());

        textField = new JTextField(10);
        textField.setFont(new Font("Arial", Font.PLAIN, 24));
        textField.setHorizontalAlignment(JTextField.RIGHT);
        textField.setEditable(false);

        JPanel panel = new JPanel(new GridLayout(4, 4));
        String[] buttonLabels = {"7", "8", "9", "+", "4", "5", "6", "-", "1", "2", "3", "*", "C", "0", "=", "/"};
        buttons = new JButton[buttonLabels.length];

        for (int i = 0; i < buttonLabels.length; i++) {
            buttons[i] = new JButton(buttonLabels[i]);
            buttons[i].addActionListener(this);
            panel.add(buttons[i]);
        }
    }
}
```



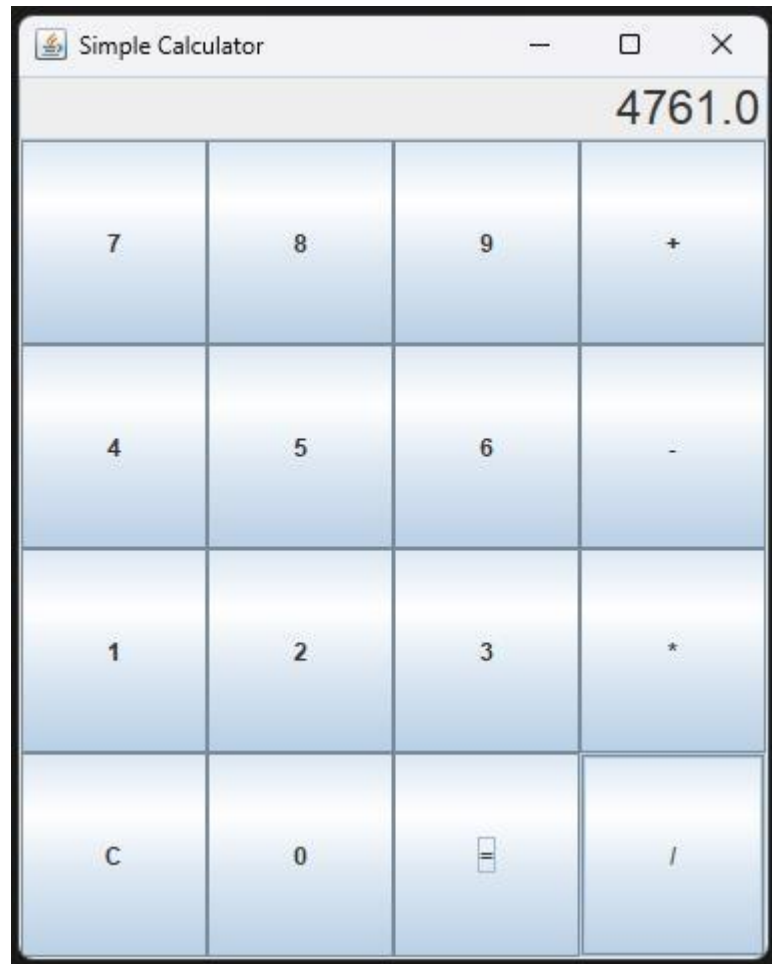
```
add(textField, BorderLayout.NORTH);
add(panel);
setVisible(true);
}

public void actionPerformed(ActionEvent e) {
String command = ((JButton) e.getSource()).getText();

if (command.matches("[0-9]")) {
    input += command;
    textField.setText(input);
}
else if (command.matches("[+\\-*/]")) {
    num1 = Double.parseDouble(input);
    operator = command.charAt(0);
    input = "";
}
else if (command.equals("=")) {
    num2 = Double.parseDouble(input);
    switch (operator) {
        case '+':
            result = num1 + num2;
            break;
        case '-':
            result = num1 - num2;
            break;
        case '*':
            result = num1 * num2;
            break;
        case '/':
            result = num1 / num2;
```

```
        break;
    }
    textField.setText(String.valueOf(result));
    input = String.valueOf(result);
}
else if (command.equals("C")) {
    input = "";
    textField.setText("");
}
}
public static void main(String[] args) {
    SwingUtilities.invokeLater(() -> new calc());
}
}
```

OUTPUT:



Source Code:

MyServer.java

```
import java.net.*;
import java.io.*;
class MyServer
{
    public static void main(String args[])throws Exception {
        ServerSocket ss=new ServerSocket(3333);
        Socket s=ss.accept();
        DataInputStream din=new DataInputStream(s.getInputStream());
        DataOutputStream dout=new DataOutputStream(s.getOutputStream());
        BufferedReader br=new BufferedReader(new InputStreamReader(System.in));
        String str="",str2="";
        while(!str.equals("stop"))
        {
            str=din.readUTF();
            System.out.println("client says:"+str);
            str2=br.readLine();
            dout.writeUTF(str2);
            dout.flush();
        }
        din.close();
        s.close();
        ss.close();
    }
}
```

MyClient.java

```
import java.net.*;
import java.io.*;
class MyClient {
    public static void main(String args[])throws Exception {
        Socket s=new Socket("localhost",3333);
        DataInputStream din=new DataInputStream(s.getInputStream());
        DataOutputStream dout=new DataOutputStream(s.getOutputStream());
        BufferedReader br=new BufferedReader(new InputStreamReader(System.in));
        String str="",str2="";
        while(!str.equals("stop"))
        {
            str=br.readLine();
            dout.writeUTF(str);
            dout.flush();
            str2=din.readUTF();
            System.out.println("Server says:"+str2);
        }
        dout.close();
        s.close();
    }
}
```

OUTPUT:

```
Client sr/bin/env /usr/lib/jvm/java-11-openjdk-amd64/bin/java -cp /home/assassinviJay73/.vscode-server/data/User/workspaceStorage/6a9f2770ff347152d04f61bf0b61955b/redhat.java/jdt_ws/htdocs_e42e8491/bin My
HELLO
Server says:HELLO
I AM CLIENT
Server says: I AM SERVER
█
```

```
Server sr/bin/env /usr/lib/jvm/java-11-openjdk-amd64/bin/java -cp /home/assassinviJay73/.vscode-server/data/User/workspaceStorage/6a9f2770ff347152d04f61bf0b61955b/redhat.java/jdt_ws/htdocs_e42e8491/bin My
HELLO
client says:HELLO
I AM SERVER
client says:I AM CLIENT
█
```

Source Code:

Client.java

```
import java.rmi.Naming;

public class Client {
    public static void main(String[] args) {
        try {
            MyRemoteInterface remoteObject = (MyRemoteInterface) Naming.lookup
                ("rmi://localhost/MyRemoteObject");

            String result = remoteObject.sendMessage("Hello from RMI Client!");
            System.out.println("Server Response: " + result);        } catch (Exception e) {
            e.printStackTrace();
        }
    }
}
```

Server.java

```
import java.rmi.Naming;
import java.rmi.registry.LocateRegistry;

public class Server {
    public static void main(String[] args) {
        try {
            MyRemoteInterface remoteObject = new ServerImple();
            LocateRegistry.createRegistry(1099);
            Naming.rebind("MyRemoteObject", remoteObject);
            System.out.println("RMI Server is running...");
        } catch (Exception e) {
            e.printStackTrace();
        }
    }
}
```

```
    }  
}  
}
```

MyRemoteInterface.java

```
import java.rmi.Remote;  
import java.rmi.RemoteException;  
  
public interface MyRemoteInterface extends Remote {  
    String sayHello() throws RemoteException;  
    String sendMessage(String message) throws RemoteException;  
}
```

ServerImple.java

```
import java.rmi.RemoteException;  
import java.rmi.server.UnicastRemoteObject;  
  
public class ServerImple extends UnicastRemoteObject implements MyRemoteInterface {  
    public ServerImple() throws RemoteException {  
    }  
  
    @Override  
    public String sayHello() throws RemoteException {  
        return "Hello from RMI Server!";  
    }  
  
    @Override  
    public String sendMessage(String message) throws RemoteException {  
        return "Message received by the server: " + message;  
    }  
}
```


OUTPUT:

```
PS C:\Users\assas\Downloads\RMI (1)> & 'C:\Program Files\Java\jdk-21\bin\java.exe' '-agentlib:jdwp=transport=dt_socket,server=n,suspend=y,address=localhost:50085' '-XX:+ShowCodeDetailsInExceptionMessages' '-cp' 'C:\Users\assas\AppData\Roaming\Code\User\workspaceStorage\8164192c3e85e7ee91753177ba6e5827\redhat.java\jdk_ws\RMI (1)_ef11b9cf\bin' 'Server'
RMI Server is running...
```

```
PS C:\Users\assas\Downloads\RMI (1)> & 'C:\Program Files\Java\jdk-21\bin\java.exe' '-agentlib:jdwp=transport=dt_socket,server=n,suspend=y,address=localhost:50099' '-XX:+ShowCodeDetailsInExceptionMessages' '-cp' 'C:\Users\assas\AppData\Roaming\Code\User\workspaceStorage\8164192c3e85e7ee91753177ba6e5827\redhat.java\jdk_ws\RMI (1)_ef11b9cf\bin' 'Client'
Server Response: Message received by the server: Hello from RMI Client!
PS C:\Users\assas\Downloads\RMI (1)>
```

Source Code:

HelloServlet.java

```
package demo;
import java.io.*;
import javax.servlet.*;
import javax.servlet.annotation.*;
import javax.servlet.http.*;

@WebServlet("/HelloServlet")
public class HelloServlet extends HttpServlet {

    // Handle both GET and POST
    protected void processRequest(HttpServletRequest request, HttpServletResponse response)
        throws ServletException, IOException {
        response.setContentType("text/html");
        PrintWriter out = response.getWriter();

        String name = request.getParameter("username");
        if (name == null || name.isEmpty()) {
            name = "Guest";
        }
        out.println("<h2>Hello, " + name + "! Welcome to Servlet.</h2>");
    }

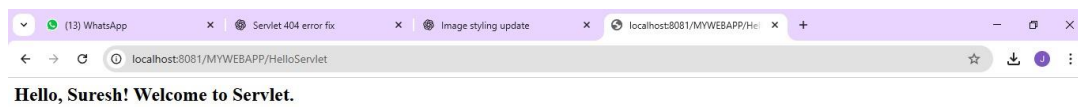
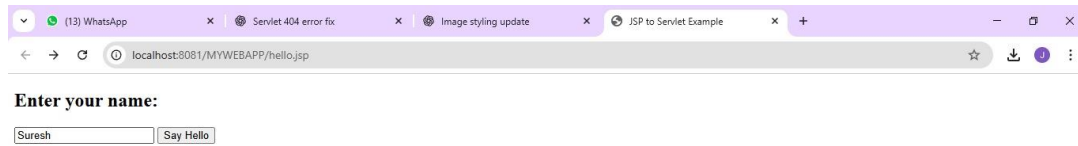
    @Override
    protected void doGet(HttpServletRequest request, HttpServletResponse response)
        throws ServletException, IOException {
        processRequest(request, response);
    }
}
```

```
@Override
protected void doPost(HttpServletRequest request, HttpServletResponse response)
    throws ServletException, IOException {
    processRequest(request, response);
}
}
```

hello.jsp

```
<!DOCTYPE html>
<html>
<head>
<title>JSP to Servlet Example</title>
</head>
<body>
<h2>Enter your name:</h2>
<form action="HelloServlet" method="post">
    <input type="text" name="username" />
    <input type="submit" value="Say Hello" />
</form>
</body>
</html>
```

OUTPUT:



Source Code:

JavaApplication2.java

```
import java.sql.Connection;
import java.sql.DriverManager;
import java.sql.ResultSet;
import java.sql.Statement;

public class JavaApplication2 {
    public static void main(String[] args) {
        Connection c = null;
        Statement stmt = null;
        try {
            Class.forName("org.postgresql.Driver");
            c = DriverManager.getConnection("jdbc:postgresql://localhost:5432/emp", "postgres",
                                           "Periyar");

            c.setAutoCommit(false);
            System.out.println("Open Database Connection Successfully");
            stmt = c.createStatement();
            ResultSet rs = stmt.executeQuery("SELECT * FROM emp_detail;");
            while (rs.next()) {
                int id = rs.getInt("eid");
                String name = rs.getString("ename");
                int age = rs.getInt("eage");
                String address = rs.getString("eaddress");
                float salary = rs.getFloat("esalary");
                System.out.println("ID = " + id);
                System.out.println("NAME = " + name);
                System.out.println("AGE = " + age);
                System.out.println("ADDRESS = " + address);
                System.out.println("SALARY = " + salary);
            }
        } catch (Exception e) {
            e.printStackTrace();
        }
    }
}
```

```
        System.out.println();
    }
    rs.close();
    stmt.close();
    c.close();
} catch (Exception e) {
    System.err.println(e.getClass().getName() + ": " + e.getMessage());
    System.exit(0);
}
System.out.println("Operation Done Successfully");
}
}
```

OUTPUT:

Open Database Connection Successfully

ID = 1

NAME = Mr.X

AGE = 30

ADDRESS = Salem, Chennai

SALARY = 50000.0

ID = 2

NAME = Mr.X

AGE = 25

ADDRESS = Salem, Chennai

SALARY = 55000.0

Operation Done Successfully

Source Code:

PersonBean.java

```
public class PersonBean {  
    private String name;  
    private int age;  
    private String address;  
    public String getName() {  
        return name;  
    }  
    public void setName(String name) {  
        this.name = name;  
    }  
    public int getAge() { return age; }  
    public void setAge(int age) {  
        this.age = age;  
    }  
    public String getAddress() {  
        return address;  
    }  
    public void setAddress(String address) {  
        this.address = address;  
    }  
}
```


JavaBean.java

```
public class JavaBean {  
    public static void main(String[] args) {  
        PersonBean person = new PersonBean();  
        person.setName("Mr.V");  
        person.setAge(21);  
        person.setAddress("SALEM ");  
        System.out.println("Name: " + person.getName());  
        System.out.println("Age: " + person.getAge());  
        System.out.println("Address: " + person.getAddress());  
    }  
}
```

OUTPUT:

```
PS C:\Users\assas\Desktop\New folder\java (1)> c:; cd 'c:\Users\assas\Desktop\New folder\java (1)'; & 'C:\Program Files\Java\jdk-21\bin\java.exe' '-agentlib:jdwp=transport=dt_socket,server=n,suspend=y,address=localhost:50360' '-XX:+ShowCodeDetailsInExceptionMessages' '-cp' 'C:\Users\assas\AppData\Roaming\Code\User\workspaceStorage\dfd99a7e5f0de3ee4429407099cad822\redhat.java\jdt_ws\java (1)_24fdad5\bin' 'JavaBean'
```

Name: Mr.V
Age: 21
Address: SALEM